

Engineering Mathematics: Fundamentals of Partial Differentiation

1. Theoretical Core: What is Partial Differentiation?

In standard calculus, a function depends on a single independent variable (e.g., $f(x)$). However, physical and engineering systems typically depend on multiple independent variables simultaneously.

Partial differentiation is the mathematical operation used to determine the rate of change of a multivariable function with respect to *one* specific variable, while holding all other independent variables strictly constant.

Mathematical Notation

The symbol ∂ (pronounced "curly d" or "partial") distinguishes partial derivatives from ordinary derivatives (d).

- $\frac{\partial}{\partial x}$: Differentiate with respect to x ; treat all other variables (like y, z) as constants.
- $\frac{\partial}{\partial y}$: Differentiate with respect to y ; treat all other variables (like x, z) as constants.
- $\frac{\partial}{\partial z}$: Differentiate with respect to z ; treat all other variables (like x, y) as constants.

2. Core Operational Rules

When executing a partial derivative, standard differentiation rules (Power Rule, Product Rule, Chain Rule) apply, supplemented by one fundamental logical constraint:

The Constant Rule of Partial Differentiation:

If $u = f(x, y)$, then when computing $\frac{\partial u}{\partial x}$, the variable y behaves exactly like a real number constant (e.g., $5, \pi, k$). Therefore:

$$\frac{\partial}{\partial x}(y) = 0$$

$$\frac{\partial}{\partial x}(y \cdot x) = y \cdot \frac{\partial}{\partial x}(x) = y \cdot 1 = y$$

3. Comprehensive Step-by-Step Problem Breakdown

Problem 1: Polynomial and Trigonometric Combination

Given the function:

$$u = x^3y + y^2x + y \sin x$$

Part A: Compute $\frac{\partial u}{\partial x}$

- **Logic:** Treat y as a constant. Apply standard derivative rules to x terms.
- **Step-by-step Expansion:**

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial}{\partial x}(x^3y) + \frac{\partial}{\partial x}(y^2x) + \frac{\partial}{\partial x}(y \sin x) \\ \frac{\partial u}{\partial x} &= y \cdot \frac{\partial}{\partial x}(x^3) + y^2 \cdot \frac{\partial}{\partial x}(x) + y \cdot \frac{\partial}{\partial x}(\sin x) \\ \frac{\partial u}{\partial x} &= y \cdot (3x^2) + y^2 \cdot (1) + y \cdot (\cos x)\end{aligned}$$

- **Final Simplified Expression:**

$$\frac{\partial u}{\partial x} = 3x^2y + y^2 + y \cos x$$

Part B: Compute $\frac{\partial u}{\partial y}$

- **Logic:** Treat x as a constant. Apply standard derivative rules to y terms.
- **Step-by-step Expansion:**

$$\begin{aligned}\frac{\partial u}{\partial y} &= \frac{\partial}{\partial y}(x^3y) + \frac{\partial}{\partial y}(y^2x) + \frac{\partial}{\partial y}(y \sin x) \\ \frac{\partial u}{\partial y} &= x^3 \cdot \frac{\partial}{\partial y}(y) + x \cdot \frac{\partial}{\partial y}(y^2) + \sin x \cdot \frac{\partial}{\partial y}(y) \\ \frac{\partial u}{\partial y} &= x^3 \cdot (1) + x \cdot (2y) + \sin x \cdot (1)\end{aligned}$$

- **Final Simplified Expression:**

$$\frac{\partial u}{\partial y} = x^3 + 2xy + \sin x$$

Problem 2: Higher-Degree Polynomial Analysis

Given the function:

$$u = x^5y^2 + 3x^2y^2 + 5xy$$

Compute $\frac{\partial u}{\partial x}$

- **Logic:** Treat y as a constant. Differentiate exclusively with respect to x .
- **Step-by-step Expansion:**

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial}{\partial x}(x^5y^2) + \frac{\partial}{\partial x}(3x^2y^2) + \frac{\partial}{\partial x}(5xy) \\ \frac{\partial u}{\partial x} &= y^2 \cdot \frac{\partial}{\partial x}(x^5) + 3y^2 \cdot \frac{\partial}{\partial x}(x^2) + 5y \cdot \frac{\partial}{\partial x}(x) \\ \frac{\partial u}{\partial x} &= y^2 \cdot (5x^4) + 3y^2 \cdot (2x) + 5y \cdot (1)\end{aligned}$$

- **Final Simplified Expression:**

$$\frac{\partial u}{\partial x} = 5x^4y^2 + 6xy^2 + 5y$$

Problem 3: Composite Function using Chain Rule (Trigonometric)

Given the function:

$$u = \sin(5x + y)$$

Compute $\frac{\partial u}{\partial x}$

- **Logic:** This is a composite function of the form $f(g(x, y))$. We must apply the **Chain Rule**: derivative of the outer function multiplied by the partial derivative of the inner function.
- **Step-by-step Expansion:**
 - Differentiate the outer function ($\sin \theta \rightarrow \cos \theta$):

$$\frac{\partial u}{\partial x} = \cos(5x + y) \cdot \frac{\partial}{\partial x}(5x + y)$$

ii. Compute the partial derivative of the inner expression $(5x + y)$ with respect to x (treating y as 0):

$$\frac{\partial}{\partial x}(5x + y) = \frac{\partial}{\partial x}(5x) + \frac{\partial}{\partial x}(y) = 5(1) + 0 = 5$$

iii. Synthesize the terms:

$$\frac{\partial u}{\partial x} = \cos(5x + y) \cdot 5$$

• **Final Simplified Expression:**

$$\frac{\partial u}{\partial x} = 5 \cos(5x + y)$$

Problem 4: Composite Function using Chain Rule (Logarithmic)

Given the function:

$$u = \log(9x + y^2)$$

Compute $\frac{\partial u}{\partial y}$

• **Logic:** Apply the Chain Rule. The derivative of the outer logarithmic function is $\frac{\partial}{\partial \theta}(\log \theta) = \frac{1}{\theta}$. Then, multiply by the partial derivative of the inner function with respect to y .

• **Step-by-step Expansion:**

i. Differentiate the outer logarithmic structure:

$$\frac{\partial u}{\partial y} = \frac{1}{9x + y^2} \cdot \frac{\partial}{\partial y}(9x + y^2)$$

ii. Compute the partial derivative of the inner expression $(9x + y^2)$ with respect to y (treating $9x$ as a constant component, which becomes 0):

$$\frac{\partial}{\partial y}(9x + y^2) = \frac{\partial}{\partial y}(9x) + \frac{\partial}{\partial y}(y^2) = 0 + 2y = 2y$$

iii. Synthesize the terms:

$$\frac{\partial u}{\partial y} = \frac{1}{9x + y^2} \cdot (2y)$$

• **Final Simplified Expression:**

$$\frac{\partial u}{\partial y} = \frac{2y}{9x + y^2}$$

4. Algorithmic Checklist for Partial Differentiation

To systematically solve any partial differential equation without error, execute these three steps:

Step	Action	Mental Framework
1	Identify the Target Variable	Look at the denominator of the operator (∂x , ∂y , or ∂z). This is your only active variable.
2	Isolate Foreign Variables	Treat all other variable symbols exactly like integers or coefficients. They do not change.
3	Apply Standard Calculus Rules	Execute standard differentiation (Power, Product, Quotient, Chain Rule). If a term contains <i>only</i> foreign variables, its derivative is unconditionally 0.